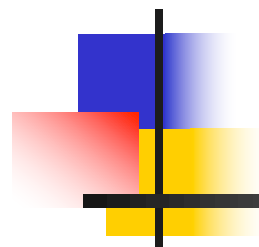
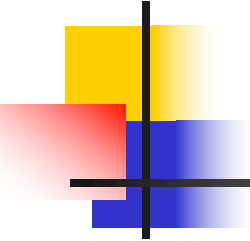


Analysis & Interpretation

Part II

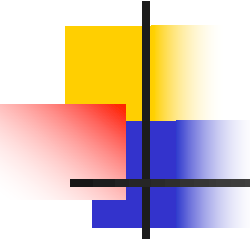


Descriptive statistics



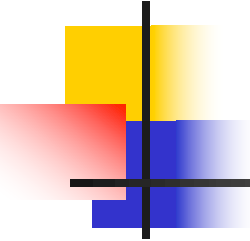
Measures of Central Tendency: Ungrouped Data

- Measures of central tendency yield information about “particular places or locations in a group of numbers.”
- Common Measures of Location
 - Mode
 - Median
 - Mean
 - Percentiles
 - Quartiles



Mode

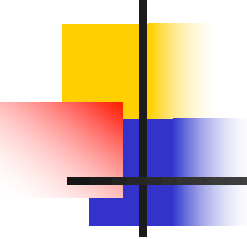
- The most frequently occurring value in a data set
- Applicable to all levels of data measurement (nominal, ordinal, interval, and ratio)
- **Bimodal** - Data sets that have two modes
- **Multimodal** - Data sets that contain more than two modes



Mode - Example

- The mode is 44.
- There are more 44s than any other value.

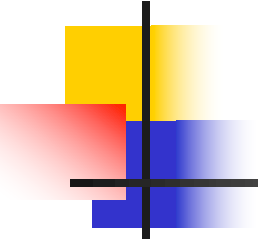
35	41	44	45
37	41	44	46
37	43	44	46
39	43	44	46
40	43	44	46
40	43	45	48



Median

- Middle value in an ordered array of numbers.
- Applicable for ordinal, interval, and ratio data
- Not applicable for nominal data
- Unaffected by extremely large and extremely small values.

Median: Computational Procedure

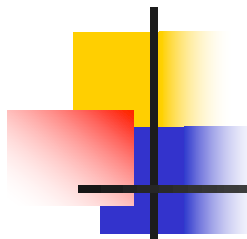


- **First Procedure**

- Arrange the observations in an ordered array.
- If there is an odd number of terms, the median is the middle term of the ordered array.
- If there is an even number of terms, the median is the average of the middle two terms.

- **Second Procedure**

- The median's position in an ordered array is given by $(n+1)/2$.



Median: Example with an Odd Number of Terms

Ordered Array

3 4 5 7 8 9 11 14 15 16 16 17 19 19 20 21 22

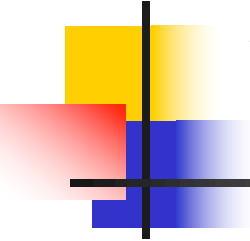
- There are 17 terms in the ordered array.
- Position of median = $(n+1)/2 = (17+1)/2 = 9$
- The median is the 9th term, 15.
- If the 22 is replaced by 100, the median is 15.
- If the 3 is replaced by -103, the median is 15.

Median: Example with an Even Number of Terms

Ordered Array

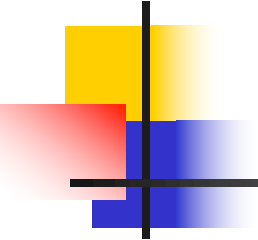
3 4 5 7 8 9 11 14 15 16 16 17 19 19 20 21

- There are 16 terms in the ordered array.
- Position of median = $(n+1)/2 = (16+1)/2 = 8.5$
- The median is between the 8th and 9th terms, 14.5.
- If the 21 is replaced by 100, the median is 14.5.
- If the 3 is replaced by -88, the median is 14.5.



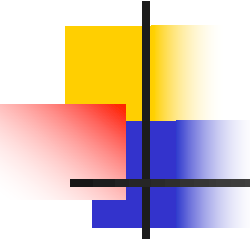
Arithmetic Mean

- Commonly called '**the mean**'
- is the average of a group of numbers
- Applicable for interval and ratio data
- Not applicable for nominal or ordinal data
- Affected by each value in the data set, including extreme values
- Computed by summing all values in the data set and dividing the sum by the number of values in the data set



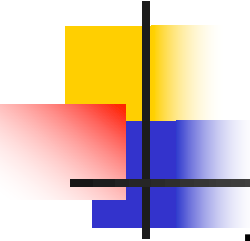
Population Mean

$$\begin{aligned}\mu &= \frac{\sum X}{N} = \frac{X_1 + X_2 + X_3 + \dots + X_N}{N} \\ &= \frac{24 + 13 + 19 + 26 + 11}{5} \\ &= \frac{93}{5} \\ &= 18.6\end{aligned}$$



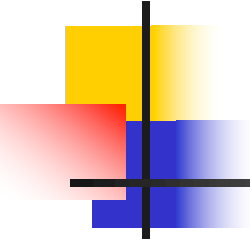
Sample Mean

$$\begin{aligned}\bar{X} &= \frac{\sum X}{n} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \\ &= \frac{57 + 86 + 42 + 38 + 90 + 66}{6} \\ &= \frac{379}{6} \\ &= 63.167\end{aligned}$$



Percentiles

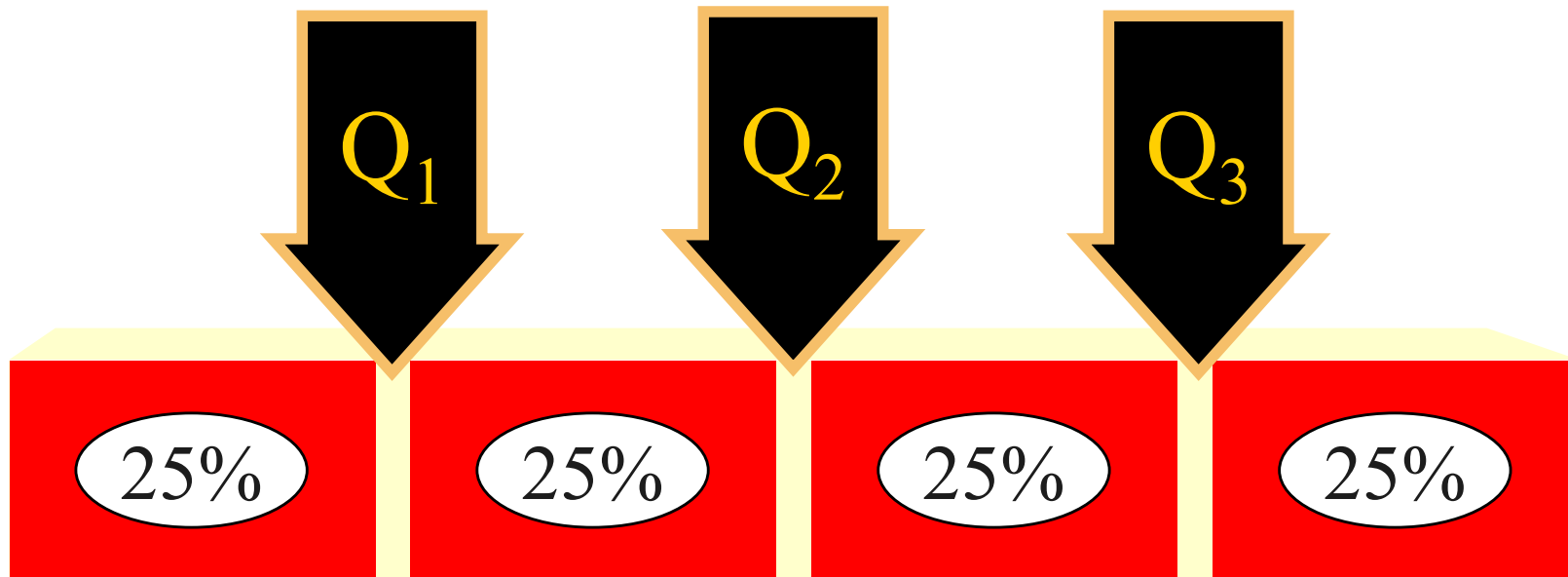
- Measures of central tendency that divide a group of data into 100 parts
- At least $n\%$ of the data lie **below the n^{th}** percentile, and at most $(100 - n)\%$ of the data lie **above the n^{th} percentile**
- **Example:** 90th percentile indicates that at least 90% of the data lie below it, and at most 10% of the data lie above it
- The median and the 50th percentile have the same value.
- Applicable for ordinal, interval, and ratio data
- Not applicable for nominal data

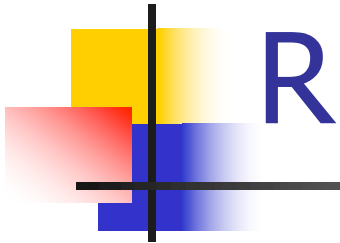


Quartiles

- Measures of central tendency that divide a group of data into four subgroups
- Q_1 : 25% of the data set is below the first quartile
- Q_2 : 50% of the data set is below the second quartile
- Q_3 : 75% of the data set is below the third quartile
- Q_1 is equal to the 25th percentile
- Q_2 is located at 50th percentile and equals the median
- Q_3 is equal to the 75th percentile
- Quartile values are not necessarily members of the data set

Quartiles

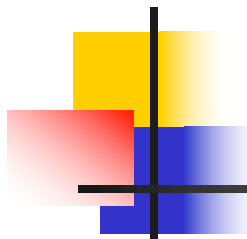




- Mode ...
- Median `median(x)`
- Mean `mean(x)`
- Quantiles `quantile(x, prob)`
1st quartile is `quantile(x, 0.25)`

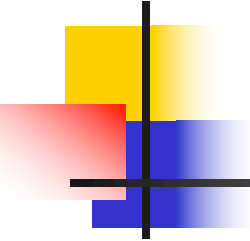
- Descriptive Statistics
`summary(x)`

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	2	3	4	5	



Measures of Variability: Ungrouped Data

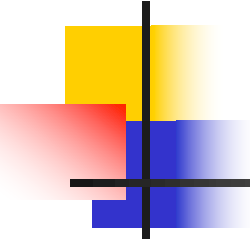
- Measures of variability describe the spread or the dispersion of a set of data.
- Common Measures of Variability
 - Range
 - Interquartile Range
 - Mean Absolute Deviation
 - Variance
 - Standard Deviation
 - Z scores
 - Coefficient of Variation



Range

- The difference between the largest and the smallest values in a set of data
- Simple to compute
- Ignores all data points except the two extremes
- **Example:**
 - Range = Largest – Smallest
 $48 - 35 = 13$

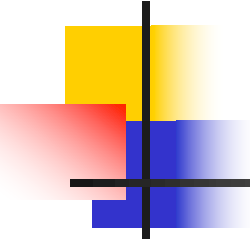
35	41	44	45
37	41	44	46
37	43	44	46
39	43	44	46
40	43	44	46
40	43	45	48



Interquartile Range

- Range of values between the first and third quartiles
- Range of the “middle half”
- Less influenced by extremes

$$\textit{Interquartile Range} = Q_3 - Q_1$$

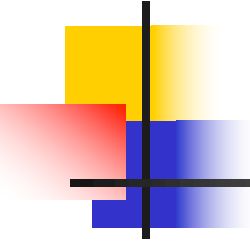


Mean Absolute Deviation

- Average of the absolute deviations from the mean

X	$X - \mu$	$ X - \mu $
5	-8	+8
9	-4	+4
16	+3	+3
17	+4	+4
18	<u>+5</u>	<u>+5</u>
	0	24

$$\begin{aligned}
 \text{M.A.D.} &= \frac{\sum |X - \mu|}{N} \\
 &= \frac{24}{5} \\
 &= 4.8
 \end{aligned}$$

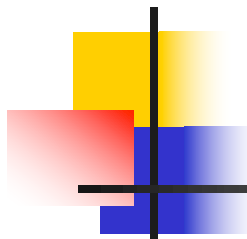


Population Variance

- Average of the squared deviations from the arithmetic mean

X	$X - \mu$	$(X - \mu)^2$
5	-8	64
9	-4	16
16	+3	9
17	+4	16
18	<u>+5</u>	<u>25</u>
	0	130

$$\begin{aligned}
 \sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\
 &= \frac{130}{5} \\
 &= 26.0
 \end{aligned}$$



Population Standard Deviation

- Square root of the variance

X	$X - \mu$	$(X - \mu)^2$
5	-8	64
9	-4	16
16	+3	9
17	+4	16
18	+5	25
	<u>0</u>	<u>130</u>

$$\begin{aligned}\sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\ &= \frac{130}{5} \\ &= 26.0 \\ \sigma &= \sqrt{\sigma^2} \\ &= \sqrt{26.0} \\ &= 5.1\end{aligned}$$

Sample Variance

- Average of the squared deviations from the arithmetic mean

X	$X - \bar{X}$	$(X - \bar{X})^2$
2,398	625	390,625
1,844	71	5,041
1,539	-234	54,756
<u>1,311</u>	<u>-462</u>	<u>213,444</u>
7,092	0	663,866

$$\begin{aligned}
 S^2 &= \frac{\sum (X - \bar{X})^2}{n - 1} \\
 &= \frac{663,866}{3} \\
 &= 221,288.67
 \end{aligned}$$

Sample Standard Deviation

- Square root of the sample variance

X	$X - \bar{X}$	$(X - \bar{X})^2$
2,398	625	390,625
1,844	71	5,041
1,539	-234	54,756
1,311	-462	213,444
<u>7,092</u>	<u>0</u>	<u>663,866</u>

$$S^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

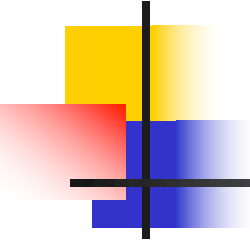
$$= \frac{663,866}{3}$$

$$= 221,288.67$$

$$S = \sqrt{S^2}$$

$$= \sqrt{221,288.67}$$

$$= 470.41$$



Empirical Rule

- Data are normally distributed (or approximately normal)

Distance from
the Mean

$\mu \pm 1\sigma$

$\mu \pm 2\sigma$

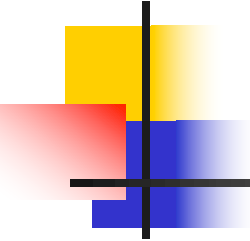
$\mu \pm 3\sigma$

Percentage of Values
Falling Within Distance

68

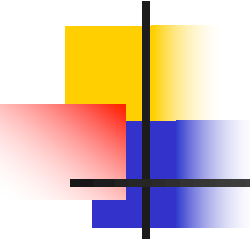
95

99.7



Empirical Rule

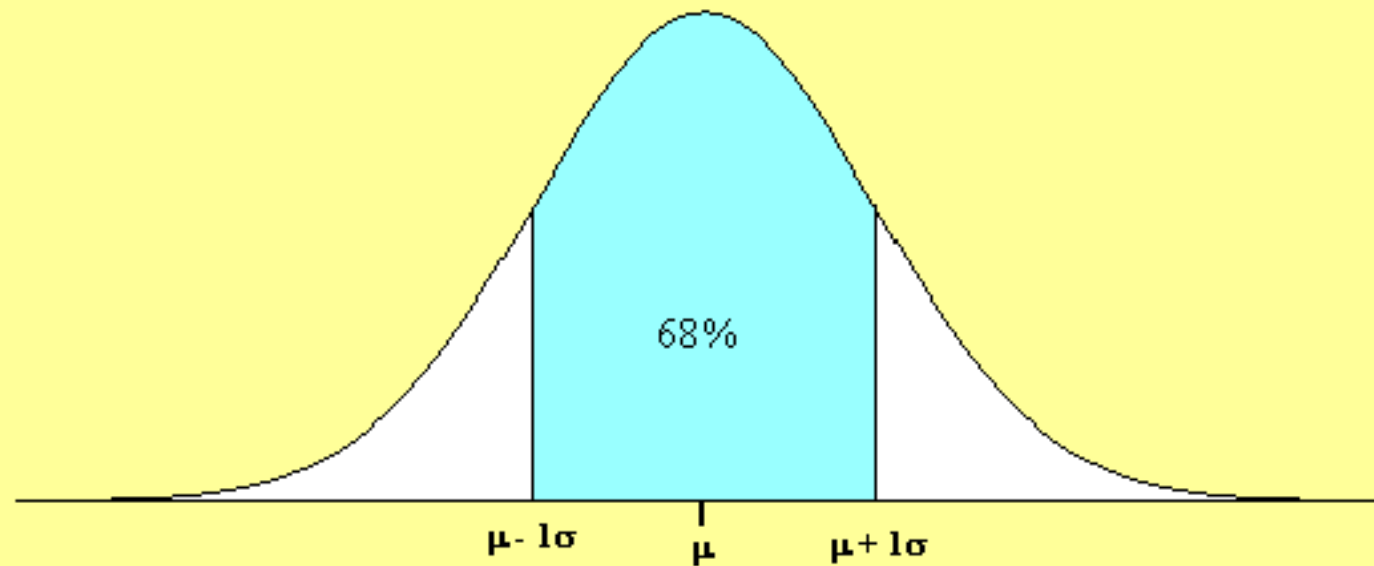
- Knowing the value of the **mean** and the value of the **standard deviation** for a data set can provide a great deal of information about the data set.
- In particular, if the data set has a single mound and is symmetrical (“**bell-shaped**”), then one can generalize some properties of the distribution.
- One such generalization is called the **Empirical Rule**.
- The Empirical Rule gives some general statements relating the mean and the standard deviation of a bell-shaped distribution.
It relates the mean to one, two, and three standard deviations.



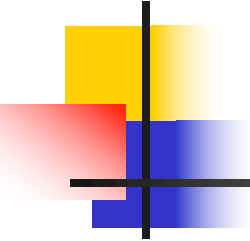
Empirical Rule – One Sigma Rule

- **One Sigma Rule** – Approximately 68% of the data values will lie within one standard deviation from the mean.
- That is, one can expect a deviation of more than one sigma from the mean to occur once in every three observations.
- This true because approximately 33% (approximately $1/3$) of the values are outside one standard deviation from the mean

Empirical Rule – One Sigma Rule



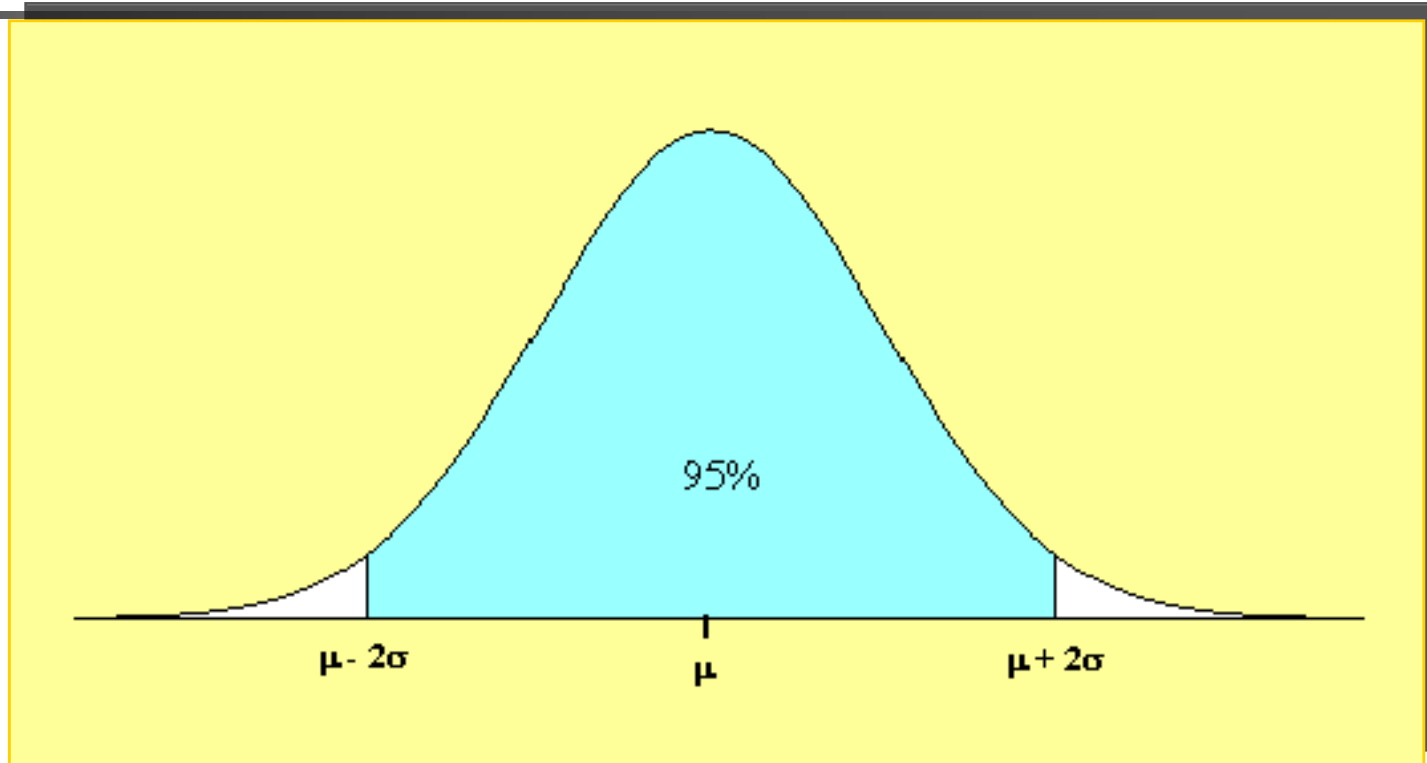
Graphical Display of the One Sigma Rule



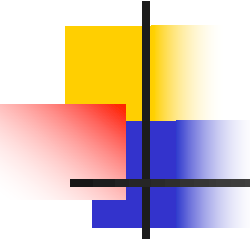
Empirical Rule – Two Sigma Rule

- **Two Sigma Rule** – Approximately 95% of the data values will lie within two standard deviations from the mean.
- That is, one can expect a deviation of more than two sigma from the mean to occur once in every twenty observations.
- This true because approximately 5% ($1/20$) of the values are outside two standard deviations from the mean

Empirical Rule – Two Sigma Rule



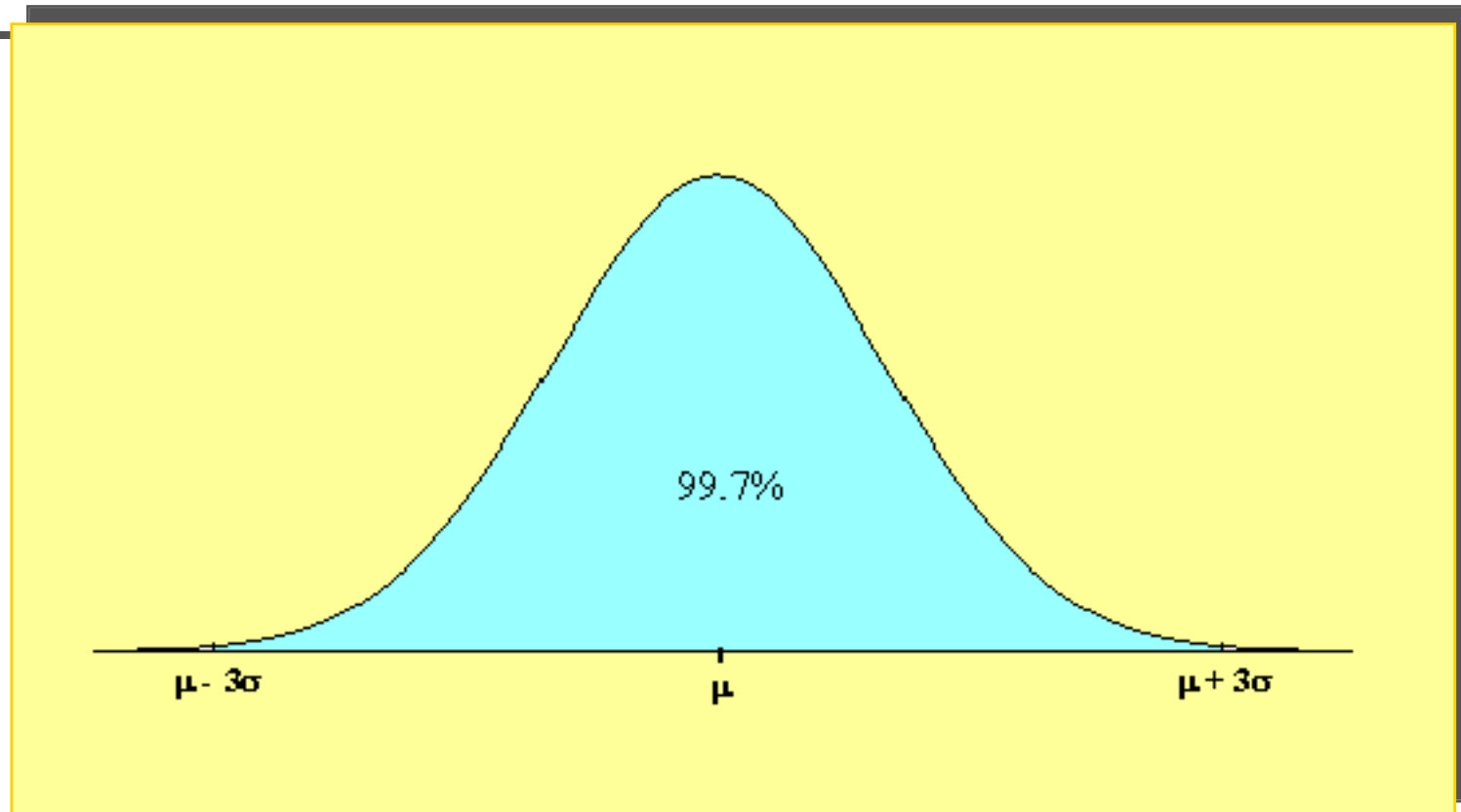
Graphical Display of the Two Sigma Rule



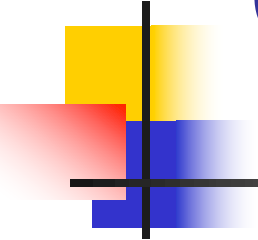
Empirical Rule – Three Sigma Rule

- **Three Sigma Rule** – Approximately 99.7% of the data values will lie within three standard deviations from the mean.
- That is, one can expect a deviation of more than three sigma from the mean to occur once in every 333 observations.
- This true because approximately 0.3% ($1/333$) of the values are outside three standard deviations from the mean

Empirical Rule – Three Sigma Rule



Graphical Display of the Three Sigma Rule



Coefficient of Variation

- Ratio of the standard deviation to the mean, expressed as a percentage
- Measurement of relative dispersion

$$C.V. = \frac{\sigma}{\mu} (100)$$



Coefficient of Variation

$$\mu_1 = 29$$

$$\sigma_1 = 4.6$$

$$C.V._1 = \frac{\sigma_1}{\mu_1}(100)$$

$$= \frac{4.6}{29}(100)$$

$$= 15.86$$

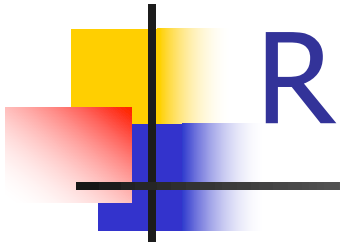
$$\mu_2 = 84$$

$$\sigma_2 = 10$$

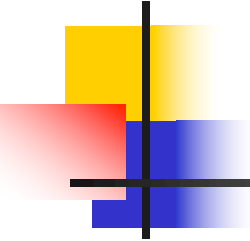
$$C.V._2 = \frac{\sigma_2}{\mu_2}(100)$$

$$= \frac{10}{84}(100)$$

$$= 11.90$$



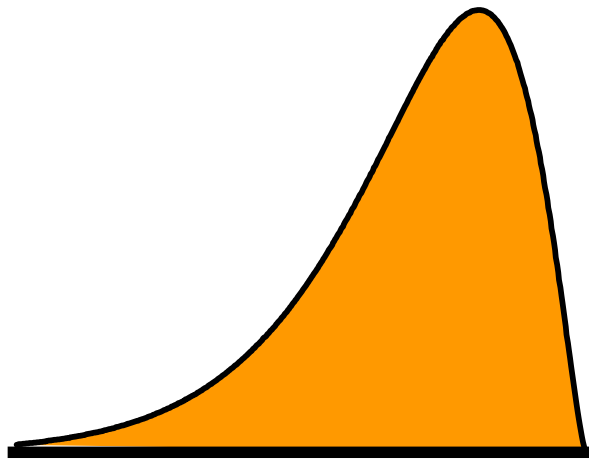
- Variance `var(x)`
- Standard deviation `sd(x)`
- Coefficient of variation `sd(x)/abs(mean(x))`



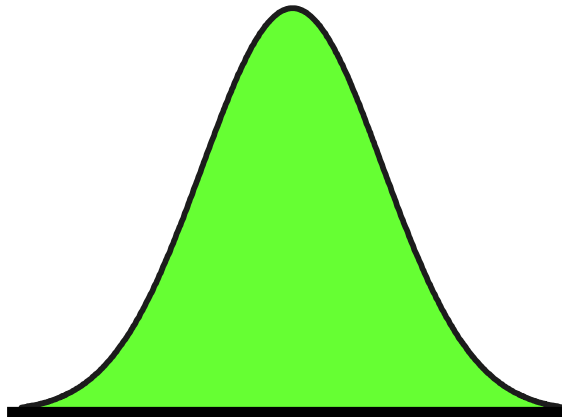
Measures of Shape

- **Skewness**
 - Absence of symmetry
 - Extreme values in one side of a distribution
- **Kurtosis**
 - Peakedness of a distribution
 - Leptokurtic: high and thin
 - Mesokurtic: normal shape
 - Platykurtic: flat and spread out
- **Box and Whisker Plots**
 - Graphic display of a distribution
 - Reveals skewness

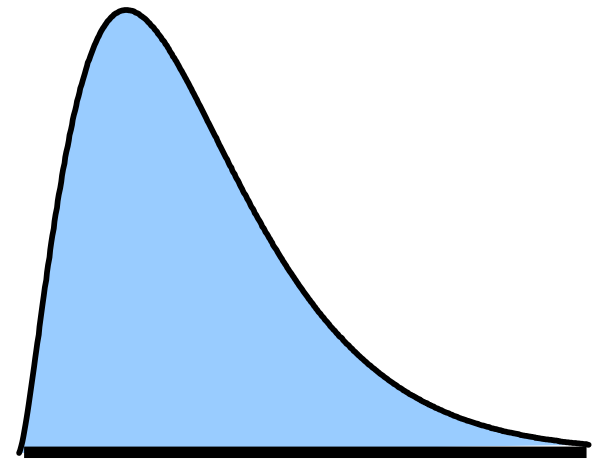
Skewness



**Negatively
Skewed**

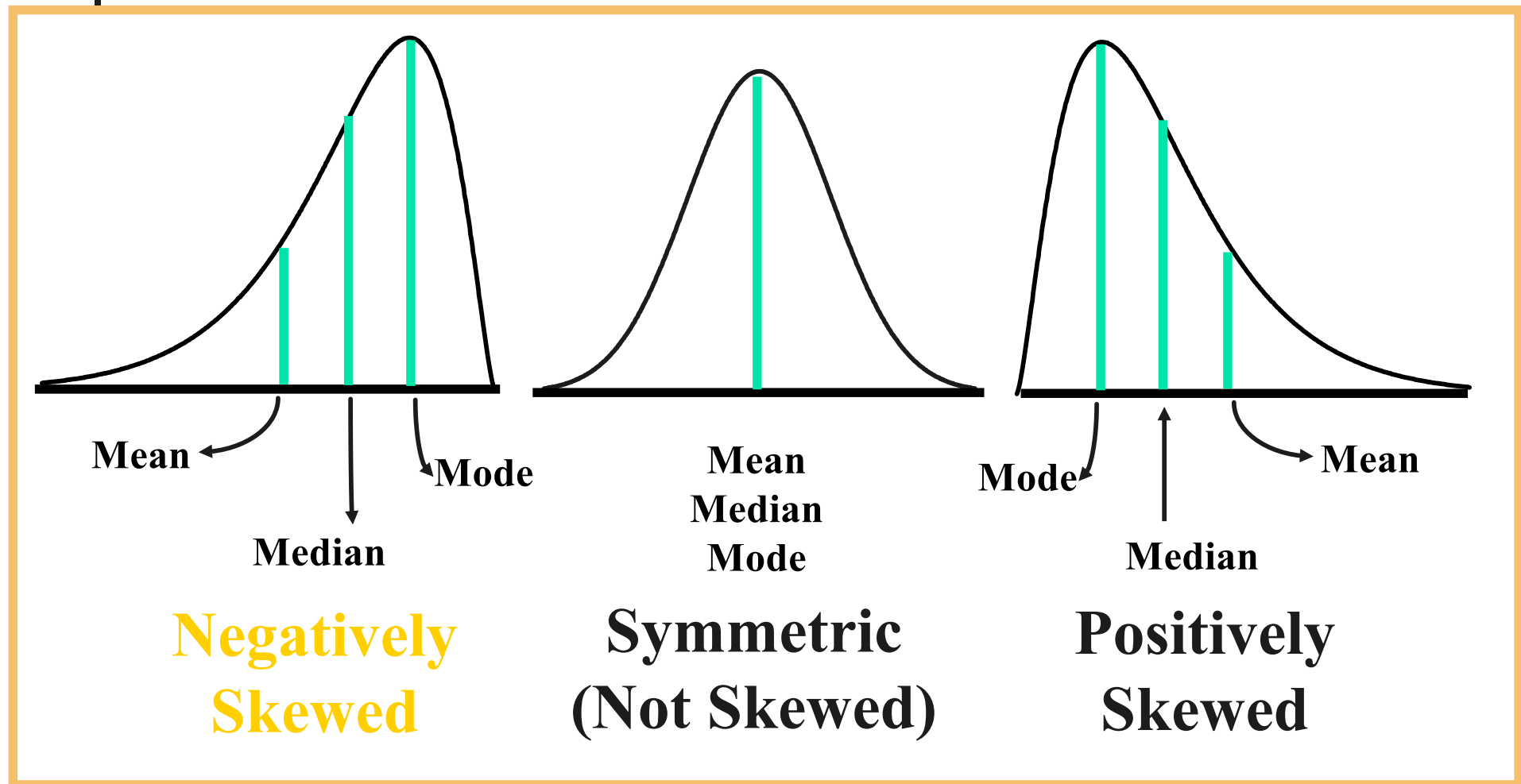


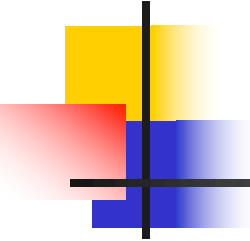
**Symmetric
(Not Skewed)**



**Positively
Skewed**

Skewness





Coefficient of Skewness

- S: Summary measure for skewness
- If $S < 0$, the distribution is negatively skewed (skewed to the left).
- If $S = 0$, the distribution is symmetric (not skewed).
- If $S > 0$, the distribution is positively skewed (skewed to the right).

R

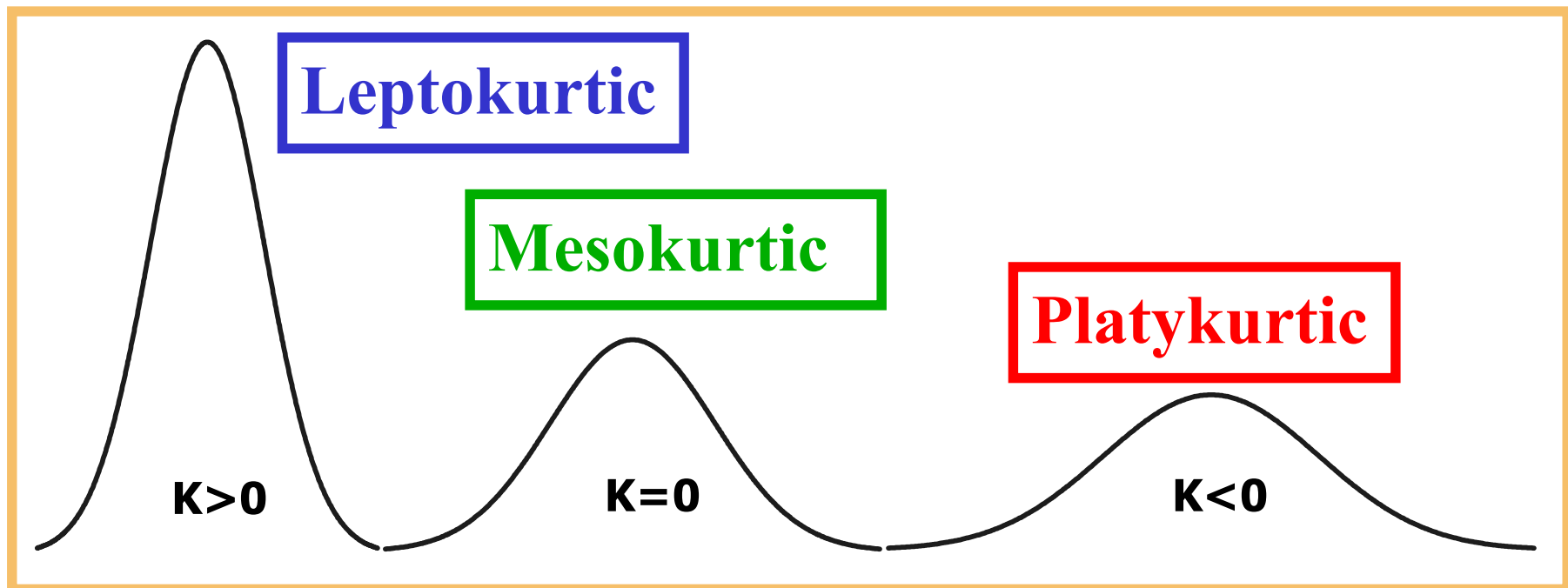
```
>library(e1071)  
>s<-skewness(x, na.rm=FALSE)
```

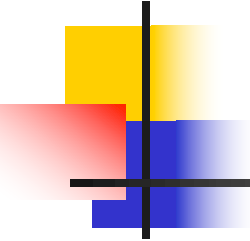
Kurtosis

- Peakedness of a distribution
 - Leptokurtic: high and thin
 - Mesokurtic: normal in shape
 - Platykurtic: flat and spread out

R

```
>library(e1071)  
>kurtosis(x, na.rm=FALSE)
```



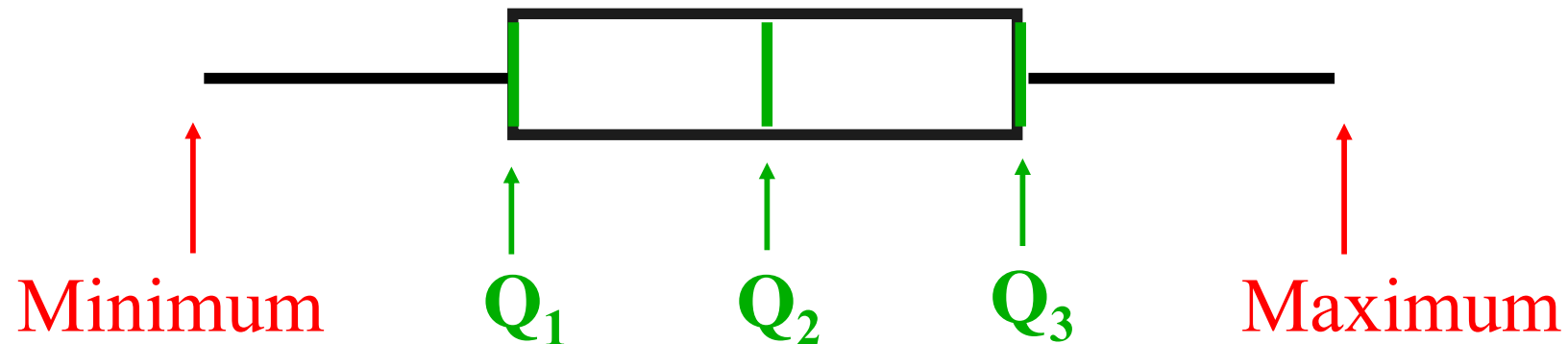


Box and Whisker Plot

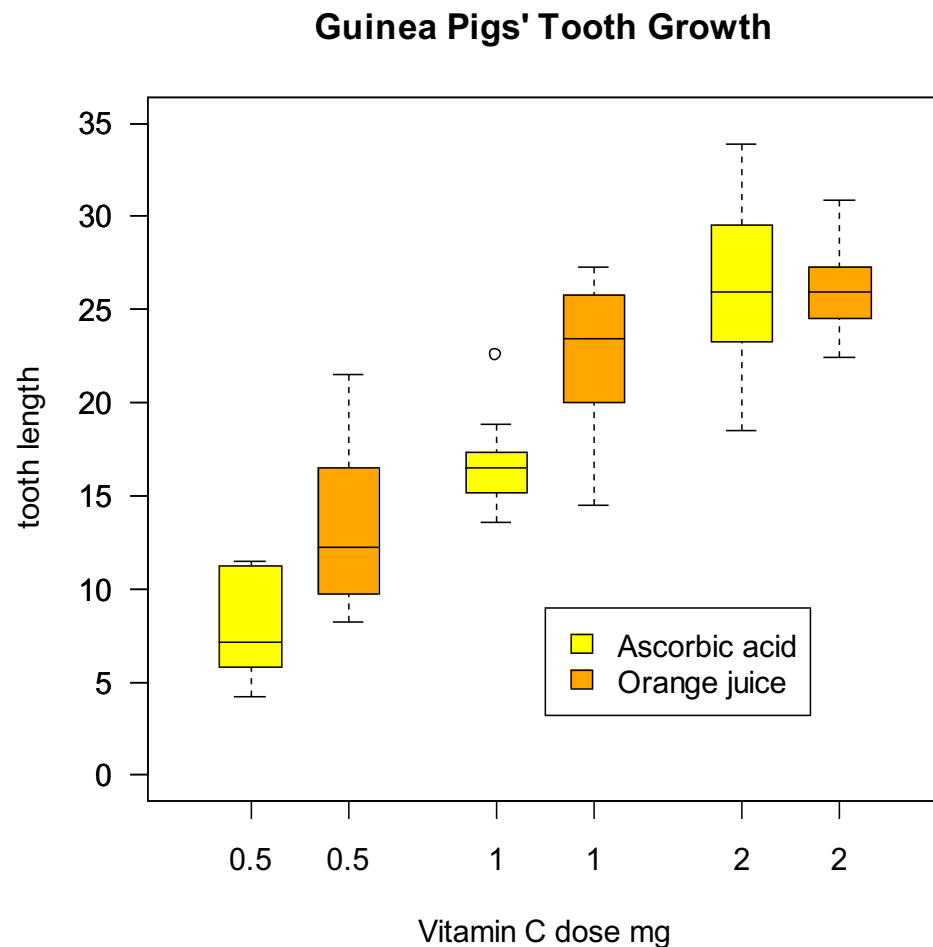
- Five values are used
 - Median, Q_2
 - First quartile, Q_1
 - Third quartile, Q_3
 - Minimum value in the data set
 - Maximum value in the data set
- Inner Fences
 - $IQR = Q_3 - Q_1$
 - Lower inner fence = $Q_1 - 1.5 IQR$
 - Upper inner fence = $Q_3 + 1.5 IQR$
 - Values outside inner fences are **mild outliers**
 - Indicated with open circles
- Outer Fences
 - Lower outer fence = $Q_1 - 3.0 IQR$
 - Upper outer fence = $Q_3 + 3.0 IQR$
 - Values outside outer fences are **extreme outliers**
 - Indicated with closed circles

```
R  
>boxplot(x)
```

Box and Whisker Plot

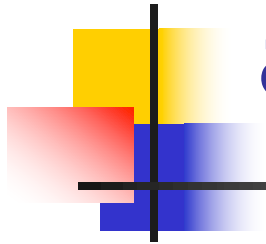


Boxplots: example

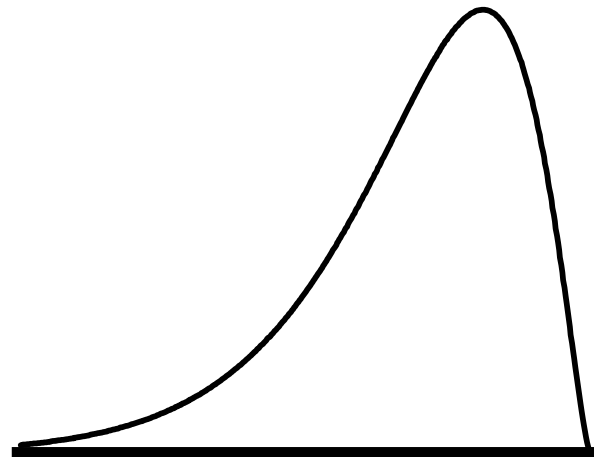


```
boxplot(t$Value~t$Cat)
```

Skewness: Box and Whisker Plots, and Coefficient of Skewness



$S < 0$



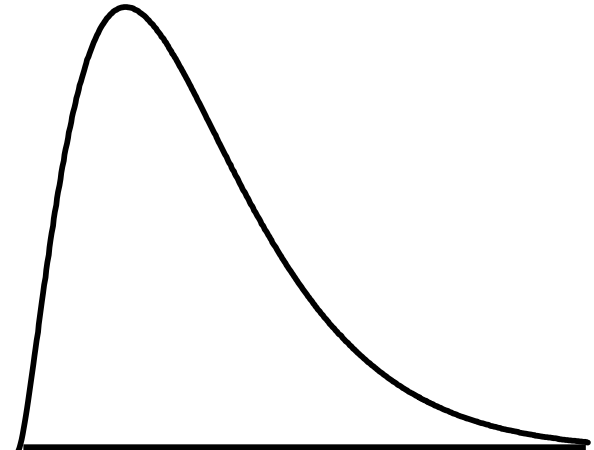
**Negatively
Skewed**

$S = 0$



**Symmetric
(Not Skewed)**

$S > 0$



**Positively
Skewed**

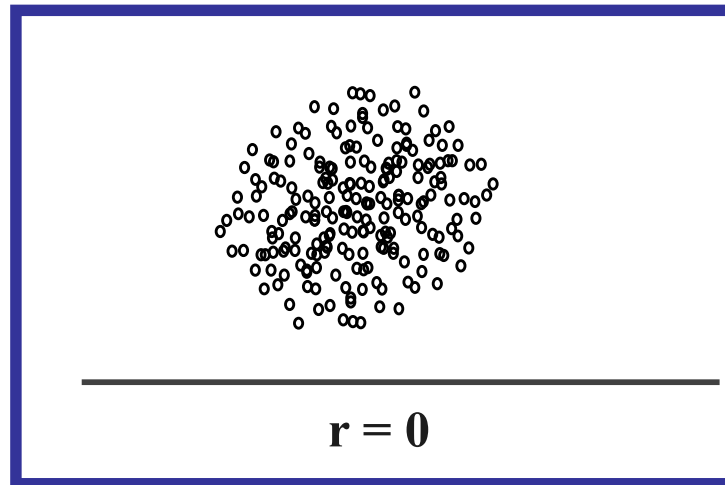
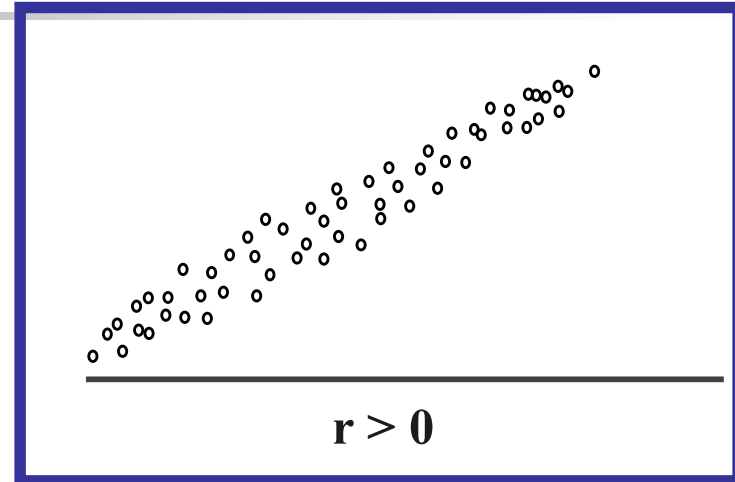
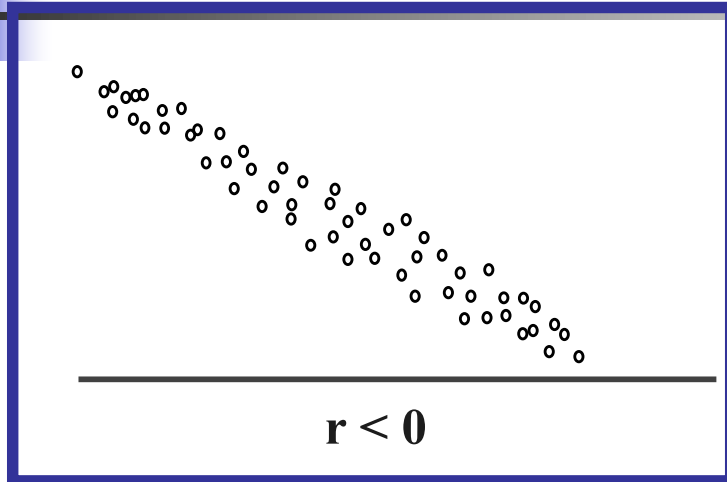
Pearson Product-Moment Correlation Coefficient

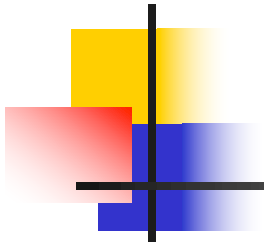
$$\begin{aligned} r &= \frac{SS_{XY}}{\sqrt{(SS_X)(SS_Y)}} \\ &= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \\ &= \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sqrt{\left[\sum x^2 - \frac{(\sum x)^2}{n} \right] \left[\sum y^2 - \frac{(\sum y)^2}{n} \right]}} \end{aligned}$$

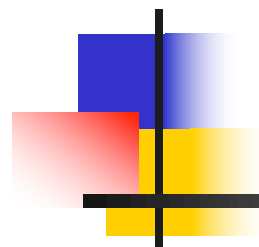
R
>cor(x,y)

$$-1 \leq r \leq 1$$

Three Degrees of Correlation



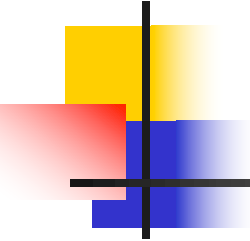




Distributions

Discrete vs Continuous Distributions

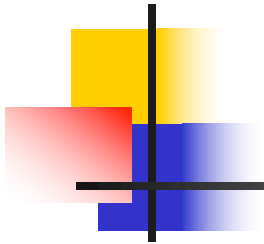
- **Random Variable** a variable which contains the outcomes of a chance experiment
- **Discrete Random Variable** the set of all possible values is at most a finite or a countably infinite number of possible values
 - Number of new subscribers to a magazine
 - Number of bad checks received by a restaurant
 - Number of absent employees on a given day
- **Continuous Random Variable** takes on values at every point over a given interval
 - Current Ratio of a motorcycle distributorship
 - Elapsed time between arrivals of bank customers
 - Percent of the labor force that is unemployed



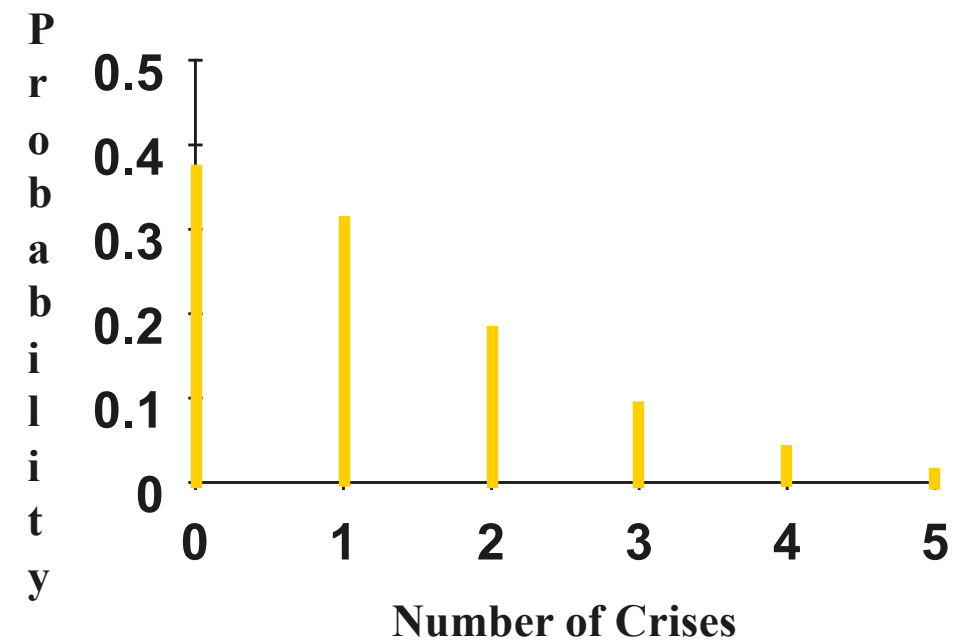
Some Special Distributions

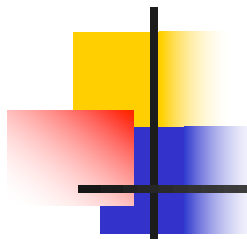
- Discrete
 - binomial
 - Poisson
- Continuous
 - normal
 - uniform
 - exponential

Discrete Distribution -- Example



Distribution of Daily Crises	
Number of Crises	Probability
0	0.37
1	0.31
2	0.18
3	0.09
4	0.04
5	0.01

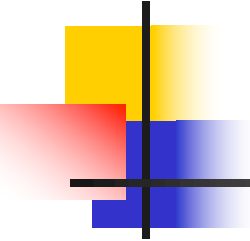




Requirements for a Discrete Probability Function

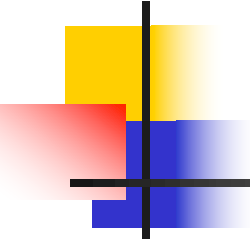
- Probabilities are between 0 and 1, inclusively $0 \leq P(X) \leq 1$ for all X
- Total of all probabilities equals 1

$$\sum_{\text{over all } x} P(X) = 1$$



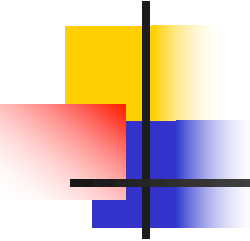
Binomial Distribution

- Experiment involves n identical trials
- Each trial has exactly two possible outcomes: success and failure
- Each trial is independent of the previous trials
 - p is the probability of a success on any one trial
 - $q = (1-p)$ is the probability of a failure on any one trial
 - p and q are constant throughout the experiment
 - X is the number of successes in the n trials



Binomial Distribution: Example

- The proportion of individuals in a random sample who support one of two political candidates fits this description
- In this case, the **statistic** is the *count* X of voters who support the candidate divided by the total number of individuals in the group n
- This provides an estimate of the **parameter** p , the proportion of individuals who support the candidate in the entire population



Binomial Distribution

- Probability function

$$P(X) = \frac{n!}{X!(n-X)!} p^X \cdot q^{n-X} \quad \text{for } 0 \leq X \leq n$$

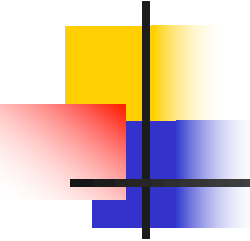
- Mean value

$$\mu = n \cdot p$$

- Variance and standard deviation

$$\sigma^2 = n \cdot p \cdot q$$

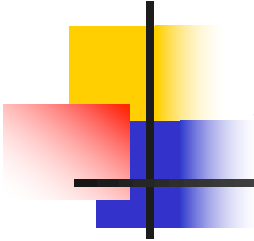
$$\sigma = \sqrt{\sigma^2} = \sqrt{n \cdot p \cdot q}$$



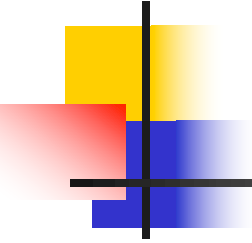
Poisson Distribution

- Describes discrete occurrences over a continuum interval
- A discrete distribution
- Describes rare events
- Each occurrence is independent any other occurrences.
- The number of occurrences in each interval can vary from zero to infinity.
- The expected number of occurrences must hold constant throughout the experiment.

Poisson Distribution: Applications



- Arrivals at queuing systems
 - airports - people, airplanes, automobiles, baggage
 - banks - people, automobiles, loan applications
 - computer file servers - read and write operations
- Defects in manufactured goods
 - number of defects per 1,000 feet of extruded copper wire
 - number of blemishes per square foot of painted surface
 - number of errors per typed page



Poisson Distribution

Probability function

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!} \quad \text{for } X = 0, 1, 2, 3, \dots$$

where:

$\lambda = \text{long-run average}$

$e = 2.718282\dots$ (the base of natural logarithms)

Mean value

λ

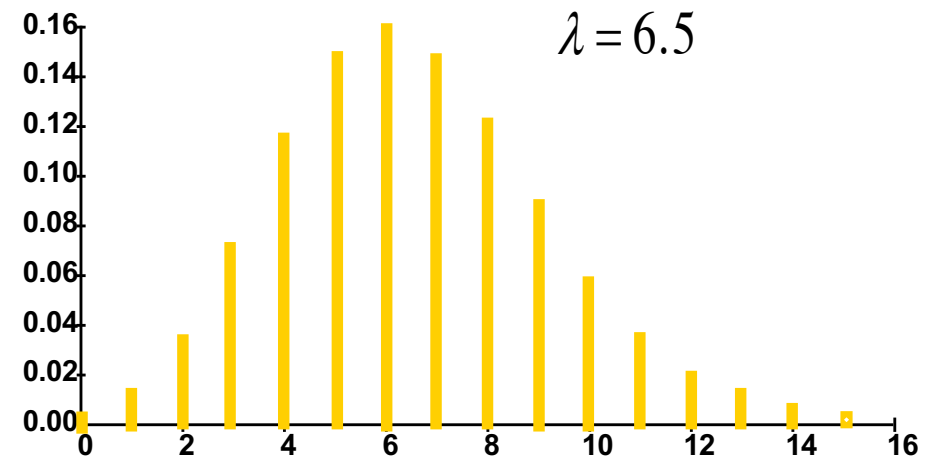
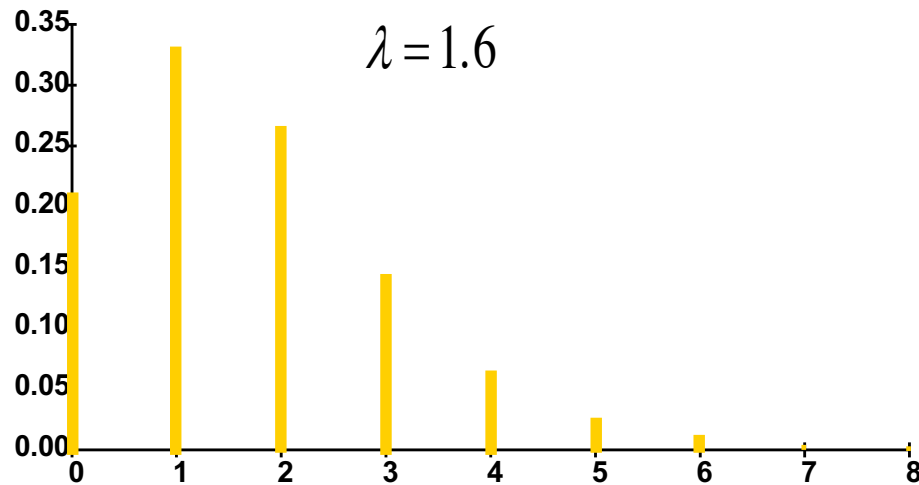
Variance

λ

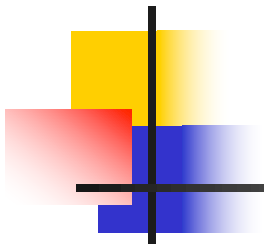
Standard deviation

$\sqrt{\lambda}$

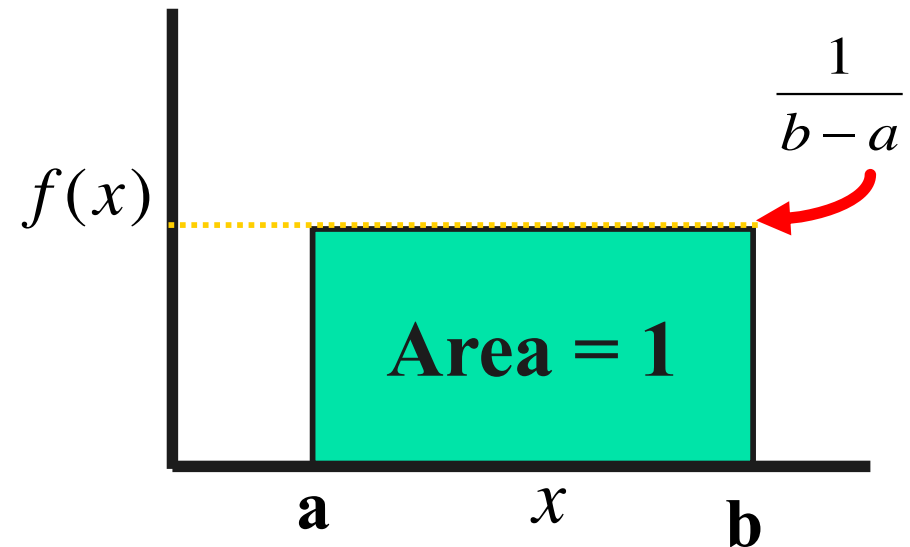
Poisson Distribution: Graphs



Uniform Distribution



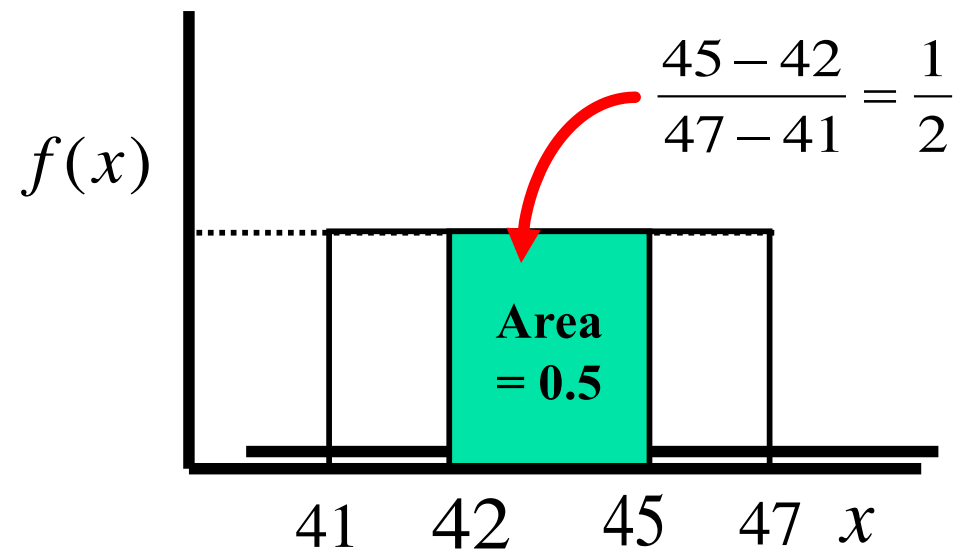
$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for all other values} \end{cases}$$



Uniform Distribution Probability

$$P(x_1 \leq X \leq x_2) = \frac{x_2 - x_1}{b - a}$$

$$P(42 \leq X \leq 45) = \frac{45 - 42}{47 - 41} = \frac{1}{2}$$



Uniform Distribution

Mean and Standard Deviation

Mean

$$\mu = \frac{a + b}{2}$$

Mean

$$\mu = \frac{41 + 47}{2} = \frac{88}{2} = 44$$

Standard Deviation

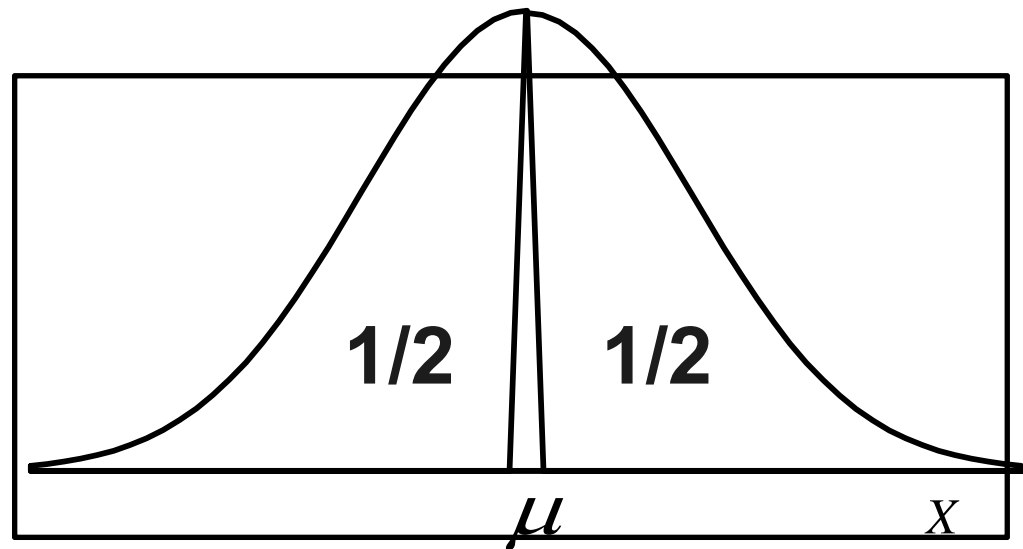
$$\sigma = \frac{b - a}{\sqrt{12}}$$

Standard Deviation

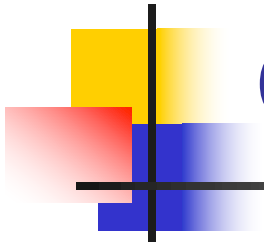
$$\sigma = \frac{47 - 41}{\sqrt{12}} = \frac{6}{3.464} = 1.732$$

Characteristics of the Normal Distribution

- Continuous distribution
- Symmetrical distribution
- Asymptotic to the horizontal axis
- Unimodal
- A family of curves
- Area under the curve sums to 1.
- Area to right of mean is $1/2$.
- Area to left of mean is $1/2$.



Probability Density Function of the Normal Distribution



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

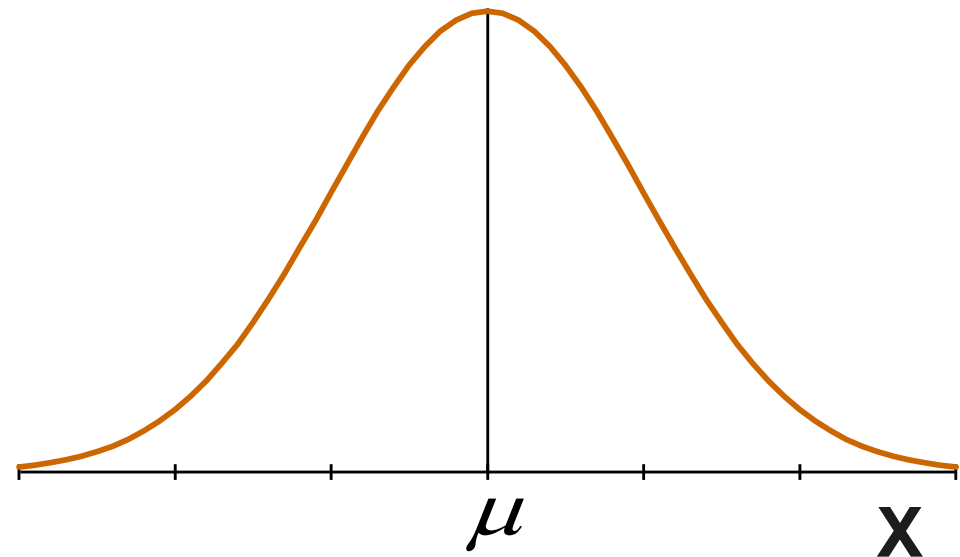
Where :

μ = mean of X

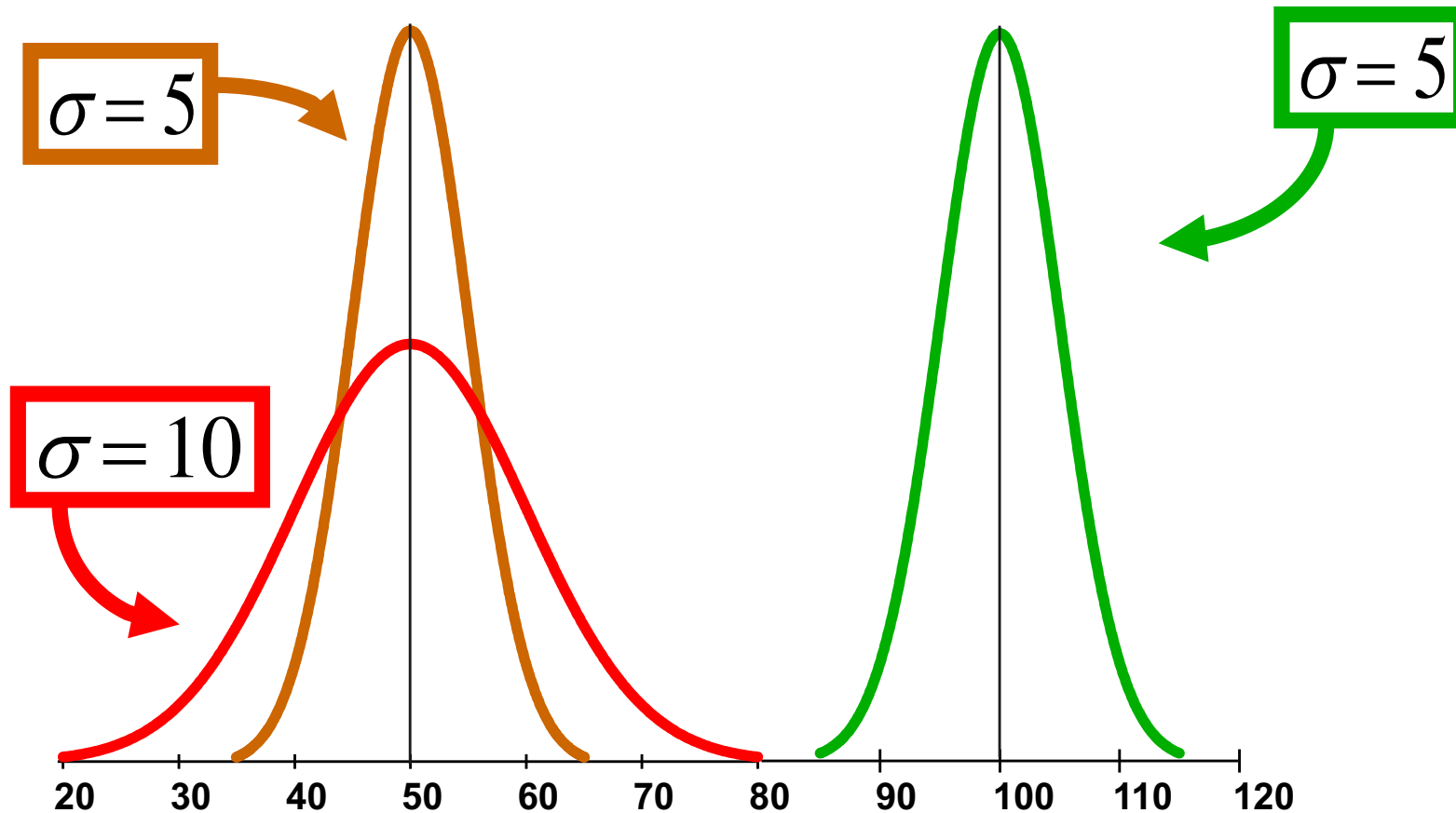
σ = standard deviation of X

π = 3.14159 . . .

e = 2.71828 . . .

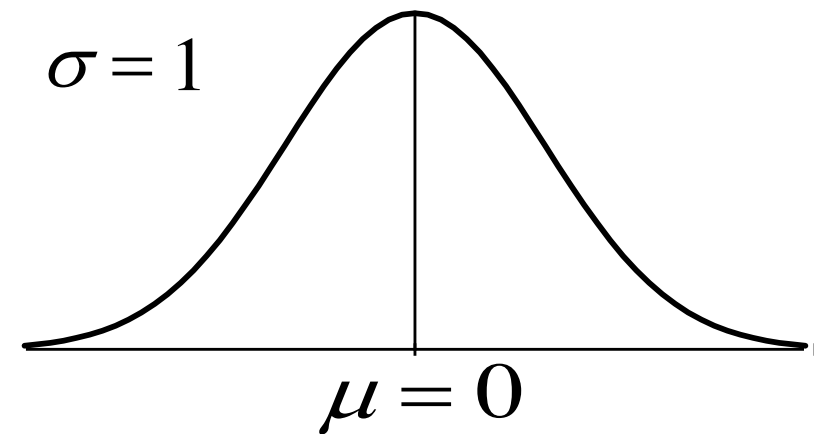


Normal Curves for Different Means and Standard Deviations

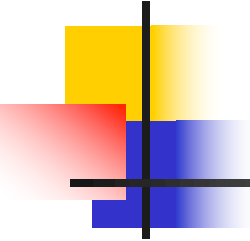


Standardized Normal Distribution

- A normal distribution with
 - a mean of zero, and
 - a standard deviation of one
- Z Formula
 - standardizes any normal distribution
- Z Score
 - computed by the Z Formula
 - the number of standard deviations which a value is away from the mean



$$Z = \frac{X - \mu}{\sigma}$$



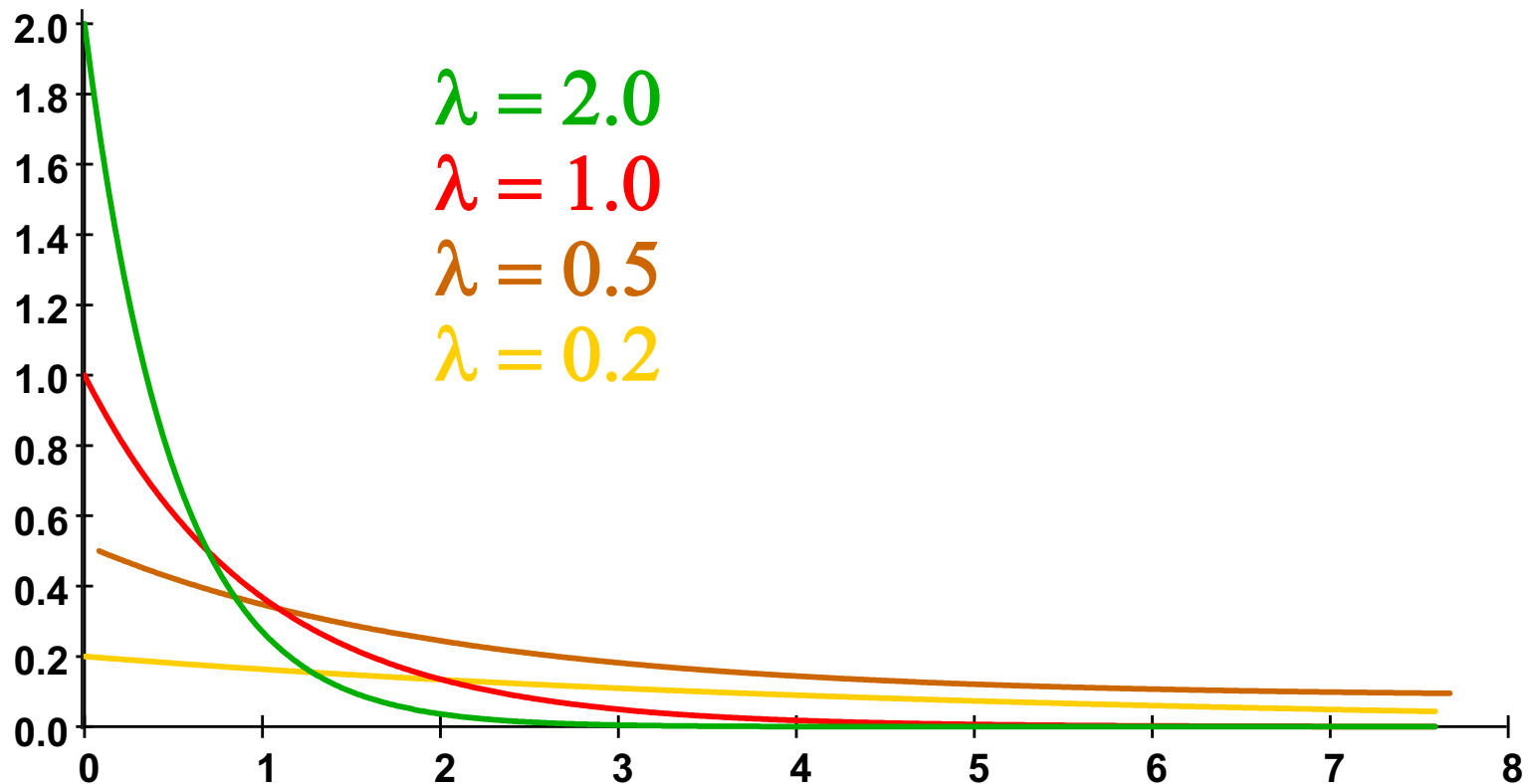
Exponential Distribution

- Continuous
- Family of distributions
- Skewed to the right
- X varies from 0 to infinity
- Apex is always at $X = 0$
- Steadily decreases as X gets larger
- Probability function

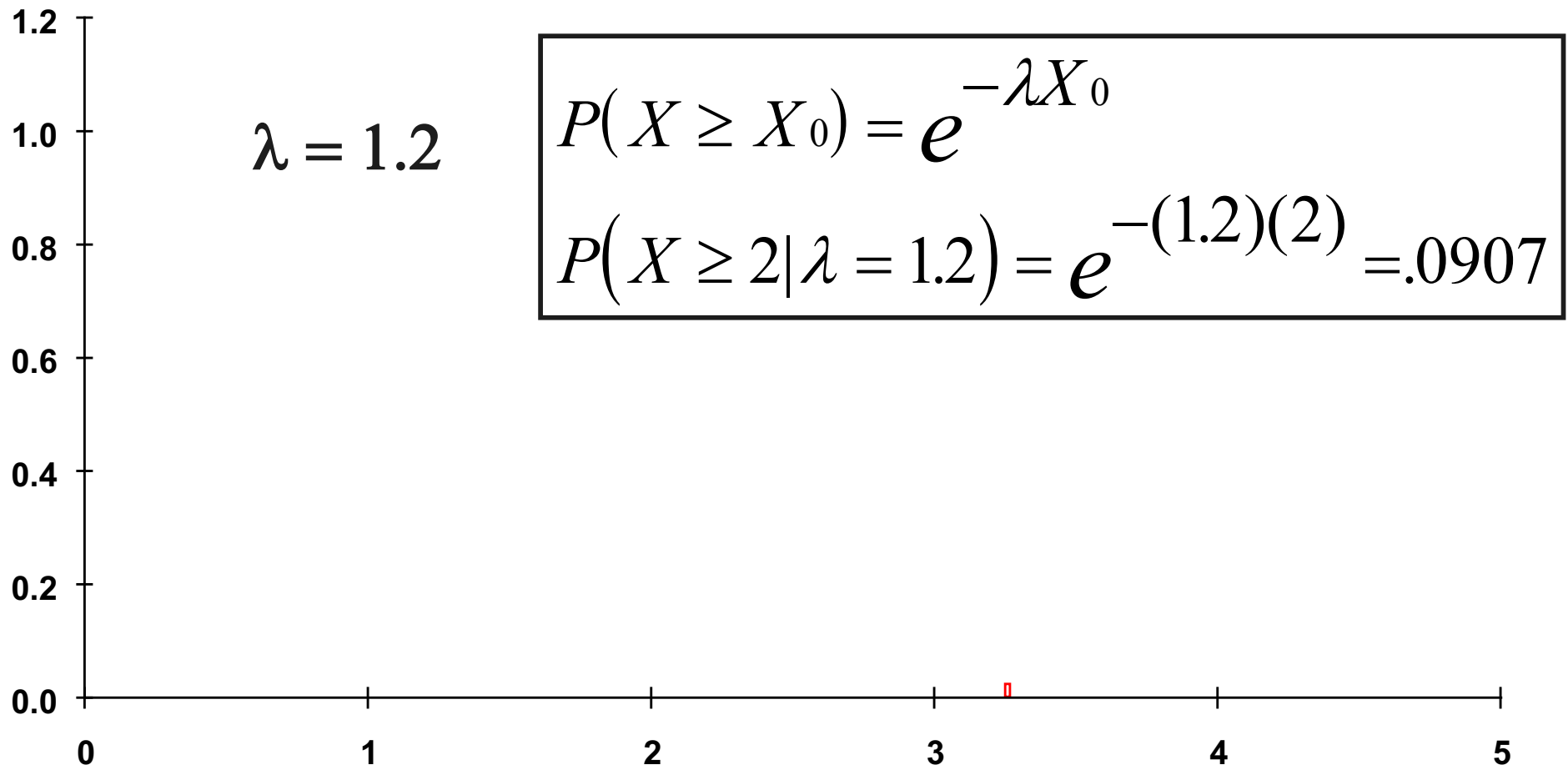
$$f(X) = \lambda e^{-\lambda X} \quad \text{for } X \geq 0, \lambda > 0$$

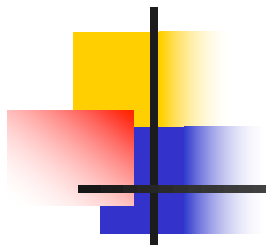
- Example: interarrival times in a queue

Graphs of Selected Exponential Distributions



Exponential Distribution: Probability Computation





R: Generating distributions

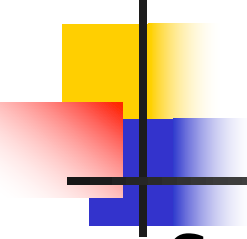
- Binomial distribution
 - `rbinom(n, size, prob)`
- Poisson distribution
 - `rpois(n, lambda)`
- Uniform distribution
 - `runif(n, min=0, max=1)`
- Normal distribution
 - `rnorm(n, mean=0, sd=1)`
- Exponential distribution
 - `rexp(n, rate = 1)`

R: Generating distribution density

- Binomial distribution
 - `dbinom(x, size, prob)`
- Poisson distribution
 - `dpois(x, lambda)`
- Uniform distribution
 - `dunif(x, min=0, max=1)`
- Normal distribution
 - `dnorm(x, mean=0, sd=1)`
- Exponential distribution
 - `dexp(x, rate = 1)`

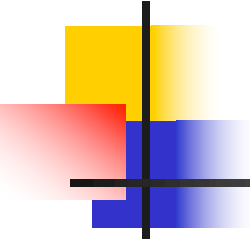


Sampling



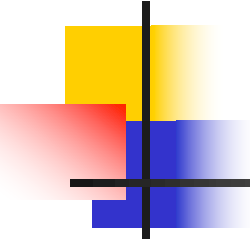
Reasons for Sampling

- Sampling can save money.
- Sampling can save time.
- For given resources, sampling can broaden the scope of the data set.
- Because the research process is sometimes destructive, the sample can save product.
- If accessing the population is impossible; sampling is the only option.



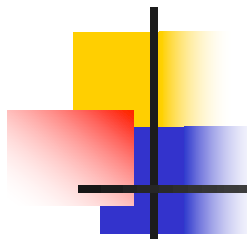
Central Limit Theorem

- For sufficiently large sample sizes ($n \geq 30$),
- the distribution of sample means $\bar{x}$, is approximately normal;
- the mean of this distribution is equal to μ , the population mean; and
- its standard deviation is $\frac{\sigma}{\sqrt{n}}$,
- regardless of the shape of the population distribution.



Central Limit Theorem

If $\bar{x}$ is the mean of a random sample of size n from a population with mean of μ and standard deviation of σ , then as n increases the distribution of $\bar{x}$ approaches a normal distribution with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

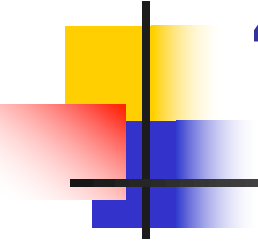


Sampling from a Normal Population

- The distribution of sample means is normal for any sample size.

If $\bar{x}$ is the mean of a random sample of size n from a normal population with mean of μ and standard deviation of σ , the distribution of $\bar{x}$ is a normal distribution with mean $\mu_{\bar{x}} = \mu$ and

standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.



Z Formula for Sample Means

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$$

$$= \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$